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Drowsiness and distraction while driving: A study based on smartphone app data

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ABSTRACT

Introduction: Due to the negative impact on road safety from driver drowsiness and distraction, several studies have been conducted, usually under driving simulator and naturalistic conditions. Nevertheless, emerging technologies offer the opportunity to explore novel data. The present study explores retrospective data, which was gathered by an app designed to monitor the driver, which is available to any driver owning a smartphone. **Method:** Drowsiness and distraction alerts emitted during the journey were aggregated by continuous driving (called sub-journey). The data include 273 drivers who made 634 sub-journeys. Two binary logit models were used separately to analyze the probability of a drowsiness and distraction event occurring. Variables describing the continuous driving time (sub-journey time), the journey time (a set of sub-journeys), the number of breaks, the breaking duration time and the first sub-journey (categorical variable) were included. Additionally, categorical variables representing the gender and age of the drivers were also incorporated. **Results:** Despite the limitations of the retrospective data, interesting findings were obtained. The results indicate that the main risk factor of inattention is driving continuously (i.e., without stopping), but it is irrelevant whether the stop is long or short as well as the total time spent on the journey. The probability of distraction events occurring during the journey is higher than drowsiness events. Yet, the impact of increasing the driving time of the journey and stopping during the journey on the probability of drowsiness is higher than the probability of distraction. Additionally, this study reveals that the elderly are more prone to drowsiness. The data also include a group of drivers, who did not provide information on gender and age, who were found to be associated to drowsiness and distraction risk. **Conclusions:** The study shows that data gathered by an app have the potential to contribute to investigating drowsiness and distraction. **Practical applications:** Drivers are highly recommended to frequently stop during the journey, even for a short period of time to prevent drowsiness and distraction.

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1. Introduction

Over the past few years, and due to a fast development of vehicle automation technology, distraction and drowsiness have received increased attention (Ferreira, Kokkinogenis, & Couto, 2019). As mentioned by Liang and Lee (2010), driver distraction has become the leading cause of motor-vehicle crashes and accounts for approximately 25% of all crash and near-crash events. Driver drowsiness is also a huge problem that is a major determinant to many traffic crashes around the world. The National Highway Traffic Safety Administration (National Center for Statistics and Analysis, 2017) estimated that in 2015, 90,000 crashes related

to sleepy drivers occurred on U.S. roadways. These crashes resulted in around 41,000 injured people and more than 800 fatalities. Due to these figures, dealing with distraction and drowsy drivers is a huge concern that requires central attention to change attitudes to improve road safety.

Distraction, fatigue, or drowsiness are examples that have led to developing Driver Monitoring Systems (DMS) implemented as standard fitments in some vehicles or as aftermarket devices. However, to date, it remains difficult to obtain quantity and quality data to allow real-time monitoring and the implementation of successful solutions available to a large public.

Due to technological advances, it has become possible to examine real-world driving using data acquired through different devices. As stated by Lenné (2013), “the widespread availability of lower cost and more advanced methods of vehicle instrumentation and recording technologies is supporting the increasing

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number of on-road research studies worldwide.” In some cases, it may be infeasible or expensive to obtain real-world data under naturalistic studies. Instead, “opportunistic” data, designated to describe retrospective data collected under real-world driving without supervision of the research team, can be an alternative. In the digital era, with the progress of the Internet and other communication systems, many possibilities have opened up, contributing to the development of new transportation data sources. The simple and constant use of smartphones, including localization systems, social media, or apps, provide the opportunity to obtain additional information alongside traditional sources. In this kind of data, the quantity and quality, as well as the space and time distribution of the information are changing all the time. These dynamic and opportunistic data depend on users’ loyalty and it cannot be controlled or verified, thus making it impossible to evaluate their veracity. Therefore, in order to take advantage of available data and gain relevant knowledge of driver behavior, a large amount of data needs to be collected to ensure interesting and representative results.

Transportation studies, including road safety analysis, are investing in this type of data collection and analysis (Blanc & Figliozi, 2016; Jestico, Nelson, & Winters, 2016; Misra, Gooze, Watkins, Asad, & Dantec, 2014). Crowdsourcing, video sharing, twitter, or other social media are examples of methods used as data sources. For instance, crowdsourced data were used to investigate the underreporting of pedestrian and bicycle crashes (Medury, Grembek, Loukaitou-Sideris, & Shafizadeh, 2017) and to analyze conflicts with bicycles at intersections (Jestico, Nelson, Potter, & Winters, 2017). Furthermore, twitter and other social media were found to be correlated to traffic incidents (Mai & Hranac, 2013; Zhang, Tang, Wang, & Wang, 2015). Additionally, Hawkins and Filtness (2017), who studied driver sleepiness on YouTube, suggested that road safety and public health should consider YouTube and humorous content as a potentially useful dissemination point of educational and interventional material.

Nevertheless, to date most of the studies focusing on distraction and drowsiness are based on naturalistic studies and/or driving simulator studies (Papantoniou, 2018; Simmons, Hicks, & Caird, 2016; Wang, Sun, Fang, Fu, & Stipancic, 2017). Although in naturalistic studies, drivers are completely free to decide their trips, both types of studies are controlled experiments, the former in real-world conditions and the latter in a simulated environment, inasmuch the research team designs the data process and the participants consent to be monitored for the purpose of the study. The diversity of study approaches can enrich knowledge about driver behavior as it is a subject in which several factors of different origins (physiological, cognitive, risk perception, road environment, vehicle, etc.) affect the results directly or indirectly, eventually leading to different findings.

In the context of emerging data, this study explores ‘opportunistic’ data obtained from a smartphone app that was developed to monitor and alert the driver when he/she becomes distracted and drowsy. Assuming distraction and drowsiness are safety-critical events, risk factors are assessed applying binary logit models separately for distraction and drowsiness. To do this, for each sub-journey (a set of sequential sub-journeys corresponds to a journey), it was established if a drowsiness/distraction event occurred (1 if occurred, 0 otherwise). The risk factors were related to the characteristics of the journey and driver. In the former, variables such as driving time, number of breaks, break duration time, and the first sub-journey were included. Additionally, the driver was characterized by gender and age.

The dataset, which included 273 drivers and 634 driving records, was obtained from a commercial DMS. In the next section, a description of the data source and the variables used is provided. The Methodology section presents the models selected to identify

and analyze the risk factors related to the occurrence of drowsiness/distraction events. The Results section shows the results of both models that are explored in depth in the Discussion section. Finally, and due to the ‘opportunistic’ character of the data, the limitations of the study and the closing remarks are clearly provided in the last section.

2. Data description

De-identified data from drivers used in this study were extracted from the DMS vendor database. The DMS was implemented by the company that sells the DMS with the aim to warn the driver rather than develop experimental research. Therefore, the research team that conducted the present study did not supervise the DMS implementation process, nor the data collection and management. Thus, the study explores ‘opportunistic’ data.

2.1. Data source

The DMS vendor is a facial biometric company operating in the automotive sector. The company’s computer-vision system aims to detect drowsiness and distraction states and warns the driver of dangerous driving behavior by sounding an alert of the same kind for both driver states.

Two different solutions were developed by the company: a fixed dashboard infrared camera to obtain images of the driver’s face and eyes and a smartphone camera-based application freely available to the general driver population. The first solution was applied to commercial drivers and already analyzed in a different perspective (Ferreira et al., 2019). In contrast, the data from the app solution comprises all kinds of drivers.

The DMS solutions used advanced techniques such as deep learning for eye state analysis and a face detection algorithm, and developments were considered for face alignment, performance with glasses, and in bad light conditions. The DMS classifies alerts as distraction or drowsiness without distinguishing the type of distraction (e.g., texting or looking outside) or drowsiness (e.g., microsleeps or fatigue mitigation). The company conducted tests by collecting videos of different subjects driving at different times of the day, in different driving situations (highway and city driving), with and without glasses. The videos were observed and the events (i.e., alerts) were verified using defined heuristics. The company indicates that the face monitoring system used in the overall system (distraction and drowsiness) has an accuracy of 93% (percentage of time the system performs well); for monitoring the eye state to detect sleepiness the company indicates an accuracy of 93% (percentage of times that correctly captured closed eyes); for monitoring the gaze to detect distraction the company indicates an accuracy of 92%. Note that tests were conducted in different conditions (night and day, different positions of the camera, with and without glasses), leading to different values, however, with a minimum value of 78.04%, for the accuracy of the eye state.

The system at the beginning and the end of each driving activity (i.e., a trip) registers a timestamp and a global position system (GPS) position. During the activity, the system emits alerts to the driver, conforming to his/her responses, and registers a series of data related to the event, such as a timestamp, a GPS position, the instant speed of the vehicle, and the type of alert (drowsiness or distraction). The data obtained from the app solution also included the driver’s age and sex as registered by the driver.

As mentioned in Ferreira et al. (2019), data treatment and rearrangement were carried out. In order to pursue the study goals, the alerts were aggregated by journey or sub-journey, and because of that, not all of the described data were used (e.g., instant vehicle speed, GPS alert).

The collected data used in this study cover the period from December 2015 to September 2016. The dataset includes 273 drivers and 634 driving records after rearranging the data to aggregate the alerts by driving records, fitting the purpose of the study. Note that the driving times less than 10 min were removed (374 cases) from the original dataset because these kinds of journey (i.e., short journeys) do not fit the purpose of the study.

2.2. Description of variables

In the studied dataset, each observation of driving records corresponds to a potential, partial journey called sub-journey. The journey may include one or more sub-journeys depending on the number of breaks. A driver ID, as well as start and end timestamps are assigned for each sub-journey. Note that the sub-journey is characterized by the driving time instead of the traveled distance because time depends on both the distance and vehicle speed, and thus, in our opinion, it represents the effect of the journey better on the driver's behavior. On the other hand, travel time is the measure usually used in past studies, thus enabling us to compare the results, particularly with the results obtained for professional drivers in (Ferreira et al., 2019).

A set of sequential sub-journeys, which correspond to the same driver, is called journey. The time between sub-journeys of the same journey is the breaking duration time. In some cases, there were high values of times between sub-journeys, suggesting that they could represent the end of the journey and the beginning of another new journey. Therefore, considering that there is no additional information about the journey, a threshold is needed to distinguish different journeys, which may include one or more sub-journeys, for the same driver. This threshold was set as the limited time interval between two consecutive sub-journeys of 9 h. This time value was based on the EC Regulation No 561/2006 from the European Parliament and the Council of 15 March, 2006 for the harmonization of certain social legislation relating to road transport, which establishes a minimum daily rest period of 9 h for professional drivers (Ferreira et al., 2019). Nevertheless, other threshold values such as 6, 8, and 10 h were tested, with no significant differences in the results. Additionally, this threshold was the same applied in Ferreira et al. (2019), and therefore, allows a comparison between the results of the studies.

In order to study distraction and drowsiness distinctly, two datasets were established by separating both types of alerts. The following variables were selected and/or created for both datasets bearing in mind that the dataset observation is a sub-journey:

- i. Sub-journey time in hours – driving time between breaks;
- ii. Journey time in hours – the cumulative driving time up to the observed sub-journey;
- iii. First sub-journey – dummy variable (0/1) in which the sub-journey is classified as 1 if it is the first of a set of sub-journeys (or the only one) and 0 otherwise;
- iv. Number of breaks – the number of breaks previous to the observed sub-journey;
- v. Breaking duration time in hours – the duration of the last break (time between sub-journeys) before the observed sub-journey;
- vi. Age – age of the driver classified by one of the following groups: 0 if between 18 and 65; 1 if equal or more than 65; 2 if unknown;
- vii. Gender – gender of the driver classified by one of the following groups: 0 if male, 1 if female and 2 if unknown.

Note that for the first sub-journey or for a journey without a break, the breaking duration time and the number of breaks are

equal to zero. Moreover, other age groups were analyzed, however, no statistical significance was found after running the models.

The database is made up of a total of 634 sub-journeys which correspond to 381 journeys made by 273 drivers. As can be observed in Table 1, the sub-journey has an average time length of 0.40 h (i.e., 24 min) for a total of 252.4 driving hours. A typical journey lasts on average 0.66 h, which is not much more than the average sub-journey time. In fact, a journey has 1.9 sub-journeys on average. The number of breaks per journey was less than 1 (see Table 1). Indeed, a significant number of the journeys (40%) were made without any breaks (i.e., with only one sub-journey). Moreover, the driver stopped around 33 min on average between driving times.

Regarding the categorical variables, the first sub-journey variable was included to eventually represent a different driver behavior in terms of inattention assuming that the driver, when starting to drive, was likely to be focused on the driving task. As shown in Table 1, 60% of the sub-journeys comprised the first sub-journey. Additionally, two categorical variables were included to represent the age and gender of the driver for each sub-journey. As can be seen in Table 1, most of the sub-journeys were driven by men (88% vs. 8% of women) aged between 18 and 65 years-old (77% vs. 9% were above 65 years-old). Finally, in 4% and 14% of the sub-journeys, there was no information about the driver's gender and age, respectively.

Regarding the alert events, in 13% of the sub-journeys, at least one alert of drowsiness was registered. In addition, in 52% of the sub-journeys, at least one alert of distraction was recorded. It can be concluded that most of the sub-journeys were driven without any driver drowsiness event, but in contrast in almost half of the sub-journeys a distraction event was detected.

3. Methodology

A binary model was used to assess the relationship between key factors and the probability associated with the occurrence of a drowsiness/distraction event. Considering a binary response and the aforementioned variables, which are gathered in vector, x , the probabilities are represented as follows:

$$\begin{aligned} \text{Prob}(Y = 1|x) &= F(x, \beta) \\ \text{Prob}(Y = 0|x) &= 1 - F(x, \beta) \end{aligned} \quad (1)$$

A set of parameters, β , reflects the impact of changes in x on the probability. A suitable model for $F(x, \beta)$ is assumed, usually considering a normal or logistic distribution. The question of which distribution to use has been widely analyzed, concluding that the two distributions tend to give similar probabilities except in the tails (Ferreira & Couto, 2012; Greene, 2008). According to Greene (2008), in most cases, the choice between these two distributions seems to make little difference. After analyzing both distributions in the present study, the logistic distribution (logit model) was selected because a small improvement in the goodness-of-fit measures was found. Therefore, a logistic distribution was used, giving rise to the following logit model:

$$\text{Prob}(Y = 1|x) = \frac{e^{\beta x}}{1 + e^{\beta x}} = \Lambda(\beta x) \quad (2)$$

where the function $\Lambda(\cdot)$ is a commonly used notation for the standard logistic distribution function.

4. Results

Two separate models called the Drowsiness Model and Distraction Model were developed using the binary logit model according to Eqs. (1) and (2). The categories called 0 of the categorical

Table 1
Descriptive statistics variables.

Continuous Variables	Mean	S.D.	Min.	Max.	Percentile 75th	Percentile 90th	Percentile 95th
Sub-journey time [hours]	0.40	0.45	0.01	4.95	0.46	0.85	1.27
Journey time [hours]	0.66	0.73	0.02	5.28	0.70	1.43	1.88
Number of breaks	0.92	1.67	0	10	1	3	4
Breaking duration time [hours]	0.55	1.46	0	8.78	0.13	2.03	4.22
Categorical variables	Category	Percentage					
First sub-journey	0: no	40					
	1: yes	60					
Age	0: 18–65	77					
	1: +65	9					
	2: unknown	14					
Gender	0: male	88					
	1: female	8					
	2: unknown	4					

variables described in Table 1 were selected as the reference category. After running the models, the results showed that the variable journey time and breaking duration time were not statistically significant in both models. Therefore, these variables were removed from the models. The variables that proved to be statistically significant, at least in one of the models, were kept in the models' results to allow a better comparison of the results of the two models, leading to the results shown in Tables 2 and 3.

As can be observed in Table 2, in the case of the Drowsiness Model results, the first sub-journey variable and the categories: Gender 1 and Age 2 are not statistically significant. The positive sign of the value estimated for the parameter of the variable Sub-journey time indicates that by increasing the driving time, the probability of a drowsiness event occurring also increases. In contrast, the negative sign of the estimated parameter of the number of breaks variable reveals that increasing the stops during the journey, the probability of drowsiness decreases. Regarding the driver's characteristics, the categorical variable shows that elderly drivers are prone to drowsiness events. Additionally, drivers, who preferred not to provide information about gender are associated to a higher probability of drowsiness than male drivers.

The results of the Distraction Model presented in Table 3 shows that the categories of Age variable and the category 1 of Gender variable are not statistically significant. Similar to the Drowsiness Model results, the estimated parameter of the Sub-journey time variable is positive and the estimated parameter of the Number of breaks variable is negative. The negative sign of the estimated parameter of the First sub-journey variable reveals that the probability of a distraction event occurring in the first sub-journey is less compared to the other sub-journeys. The model results indicate

Table 3
Results of Distraction Model.

Explanatory variable	Parameter estimate	Standard error	$P[Z > z]$
Constant	0.6044	0.2639	0.0220
Sub-journey time	0.7241	0.2296	0.0016
Number of breaks	−0.5504	0.1159	0.0000
First sub-journey	−0.5612	0.24715	0.0387
Age 0	–	–	–
Age 1	−0.1465	0.2990	0.6242
Age 2	−0.0789	0.2587	0.7605
Gender 0	–	–	–
Gender 1	0.4141	0.3148	0.1883
Gender 2	−1.1803	0.5003	0.0183

Log likelihood function: −401.26653.
Restricted log likelihood: −439.13980.
McFadden Pseudo R -squared: 0.0862442.
Hosmer-Lemeshow chi-squared: 16.01500.
Efron's R -squared: 0.114.

that the driver's age is not associated to the probability of a distraction event occurring, however drivers that did not provide the gender information are prone to distraction events occurring.

5. Discussion

This study explores 'opportunistic' data, which were obtained from a DMS available to any driver owning a smartphone, in order to analyze factors that influence the probability of a drowsiness and distraction event occurring. The studied factors are related to the journey and driver's characteristics. Despite the fact that various studies have been developed to study driver behavior focusing on drowsiness and distraction, to the best of our knowledge, there are no similar studies to the present one. Additionally, most of the past studies used a driver monitoring system that recorded the driver state but did not alert the driver, and therefore having no impact on the behavior. As mentioned by Fitzharris, Liu, Stephens, and Lenné (2017), when drowsiness feedback was provided to the driver, the number of events decreased significantly. In fact, when feedback is given to the driver (i.e., sounding an alert in this case), the driver can decide to stop and have a rest or a coffee, for example, and, as such, alters the journey concerning the situation in which there is no feedback. Therefore, no direct comparison of findings from other studies can be provided. Nevertheless, the present findings can be related to conclusions from other studies.

In this study, two models were considered to analyze distraction and drowsiness separately. Indeed, the results show some effect differences of the variable on the probabilities of drowsiness and distraction occurring. This is particularly true in the case of variables describing the drivers' age and gender. The probability of a drowsiness event occurring is highest for older drivers and

Table 2
Results of Drowsiness Model.

Explanatory variable	Parameter estimate	Standard error	$P[Z > z]$
Constant	−1.7705	0.4825	0.0002
Sub-journey time	0.70043	0.2326	0.0026
Number of breaks	−0.6997	0.2868	0.0147
First sub-journey	−0.5556	0.4931	0.2599
Age 0	–	–	–
Age 1	1.1233	0.3675	0.0022
Age 2	0.3288	0.3532	0.3519
Gender 0	–	–	–
Gender 1	−0.3796	0.5221	0.4673
Gender 2	2.2970	0.4816	0.0000

Log likelihood function: −216.51419.
Restricted log likelihood: −244.16954.
McFadden Pseudo R -squared: 0.1132629.
Hosmer-Lemeshow chi-squared: 11.26487.
Efron's R -squared: 0.098.

in contrast, the driver's age did not show any effect on the probability of a distraction event occurring. On the other hand, the drivers who did not provide gender information (category 2) are related to the probabilities of drowsiness and distraction, however with opposite effects. Apparently, these drivers represent a risk group in terms of drowsiness and, in contrast, they are associated to a lower risk of distraction when compared to male drivers. Unfortunately, no practical guidance can be provided given the content of the information.

Despite the fact that no comparison can be made with other studies and even among past studies, which also investigated the impact of age, because there are several differences regarding the experimental conditions and age groups that may explain the different results, we have summarized some interesting past findings. For instance, [Filtiness, Reyner, and Horne \(2012\)](#) mentioned that following a normal sleep, no differences between age groups were found during simulated driving experiments. Even after sleep restriction, younger drivers showed significantly more sleepiness-related measures, leading to the conclusion that younger drivers have greater vulnerability to sleep loss when driving for a long time in the afternoon. In addition, a crash analysis conducted by [Tefft \(2010\)](#) showed that younger drivers aged 16–24 were nearly twice as likely to be involved in a drowsy driving crash as drivers aged 40–59. In the case of driver distraction, different kinds of conclusions were found eventually depending on the type of analyzed distraction. For instance, [Strayer and Drews \(2004\)](#) found that the driving performance of both younger and older adults was influenced by cell phone conversations with equivalent effects. Additionally, [Guo et al. \(2017\)](#) concluded that visual–manual distractions impact drivers of all ages, whereas cognitive distraction may have a larger impact on young drivers. Along the same lines, [Pope, Bell, and Stavrinou \(2017\)](#) reported that distracted driving is a ubiquitous phenomenon evident in drivers of all ages. The gender role was also analyzed in various studies. In the case of drowsiness, [Tefft \(2010\)](#) found that two out of three drivers involved in a drowsy driving crash were men. Curiously, [Obst, Armstrong, Smith, and Banks \(2011\)](#) concluded, based on a survey, that risk taking behavior with regards to driving while sleepy occurred in both male and female drivers and across all age groups. However, young to middle aged drivers and male drivers reported the highest frequency of driving while sleepy and reported the lowest perceived personal risk regarding driving while sleepy.

In the case of distraction, few studies were found that focused on the gender issue of the driver, however in the ESRA project ([Trigoso, Areal, & Pires, 2016](#)) where an online questionnaire was conducted at European level, it was found that women and older people are more aware of the risks of using a mobile phone while driving and do it less. [Green et al. \(2007\)](#) conducted a naturalistic driving study (96 drivers, 136,792 miles) under the SAVE-IT project to determine how distracted and normal driving differ. The study results revealed that men were more distracted (56% of the frames) than women (44%).

Considering the variables describing the journey, similar trends between drowsiness and distraction can be set out based on both models' results. It seems clear that increasing the continuous driving time, the probabilities of either a distraction or drowsiness occurring also increase. These findings are supported by several studies particularly in the case of drowsiness research in which the time length of the driving task is always a research issue, especially in the case of professional drivers ([Anund, Ihlström, Fors, Kecklund, & Filtiness, 2016](#); [Di Milia & Kecklund, 2013](#); [Mahajan, Velaga, Kumar, Choudhary, & Choudhary, 2019](#); [Pylkkönen et al., 2015](#)). Indeed, a previous study focusing on professional drivers, in which similar data obtained from a DMS were used ([Ferreira et al., 2019](#)), clearly shows this effect. Nevertheless, other studies on non-professional drivers were conducted such as a study by [Van Der Hulst, Meijman, and Rothengatter \(2001\)](#) in which a driv-

ing simulator was used showing that fatigue, which was associated with deterioration of perceptual-motor performance, increased over time. Moreover, [Wang, Bie, and Li \(2016\)](#) found that when driving time is increased, the average heart rate of drivers increases as well, implying that the active fatigue level of the driver becomes worse. In the case of driver distraction, no specific studies focusing on this variable were found in a simple search. Nevertheless, [Iseland, Johansson, Skoog, and Dåderman \(2018\)](#) reported that long-haul truck drivers feel bored and use secondary tasks as a coping strategy to alleviate boredom/drowsiness, and for social interaction. These secondary tasks are described as work environment related “necessities” (e.g., getting food and/or beverages from the refrigerator/bag, eating, drinking, removing a jacket, rubbing their face, and adjusting the seat), interacting with a mobile phone/in-truck technology, and doing administrative tasks. Overall, driving time can be assumed as an exposure variable to the risk of drowsiness and distraction occurrence. On the other hand, if the driver stops, and regardless of why he/she stopped, the probabilities of drowsiness and distraction occurrence decrease. Again, regarding driver distraction, no particular study was found focusing on the effect of a journey break on this driver state, however in the case of drowsiness, studies have been published particularly focusing on professional drivers, showing the same kind of results ([Bunn, Slavova, & Rock, 2017](#); [Ferreira et al., 2019](#); [Mahajan et al., 2019](#)).

Overall, the findings of the present study related to the sub-journey time and number of breaks variables are in agreement with the idea of the load of the driving task. Interestingly, the journey time (i.e., the total driving time) and the break duration time did not show any effect on both driver states, leading to the conclusion that the continuous driving time and stopping during the journey are the main factors contributing to the phenomenon. It is interesting to note that the first sub-journey affects the driver distraction differently, but with no impact on drowsiness. In the former, the probability of a distraction event occurring during the first sub-journey is lower than in the other following sub-journeys. It seems that, in the case of distraction, the driver is more concentrated on the driving task during the first sub-journey, and after that the driver starts to be more prone to other tasks, maybe because he/she has become bored. In contrast, drowsiness seems to be a driver behavior mostly affected by the continuous driving time, either represented by the sub-journey time and number of breaks, regardless of the sub-journey sequence.

In order to better understand the magnitude of the effects of the driving time and number of breaks variables, the probabilities were computed for different scenarios assuming the mean values of each variable (see [Table 1](#)) and the 0 category for the categorical variables for the reference scenario. The partial effect is computed by comparing the probabilities of the analyzed scenario (S_a) with the reference scenario (S_r), i.e. $(S_a - r)/S_r$. Two scenarios, besides the reference scenario, are analyzed and described in [Table 4](#): one assuming an increase of 20% of the sub-journey time and the other considering one stop instead of non-stop.

As can be seen in [Table 4](#), the partial effect of the 20% increase of driving time is highest in the case of drowsiness. Moreover, the break during the journey shows a much higher partial effect on drowsiness than distraction. Note, however, that the probabilities associated with distraction are higher. Finally, the partial effects lead to the conclusion that the number of breaks has a great reduction effect on both distraction and drowsiness.

6. Limitations and closing remarks

The exploration of “opportunistic” data is clearly advantageous, such as easy access to information and reaching a wide range of

Table 4
Probabilities of drowsiness and distraction events.

Scenario	Drowsiness		Distraction	
	Probability (%)	Partial effect (%)	Probability (%)	Partial effect (%)
Reference	18.4	–	71.0	–
Increase 20% of sub-journey time	19.2	+4.7	72.2	+1.7
1 stop	10.1	–45.3	58.5	–17.6

contributions. However, several limitations are associated to this sampling process. The most relevant concern consists of the impossibility of access and observing data origin, resulting in retrospective and uncontrolled information. Additionally, journey schedule and purpose, driver characteristics, or real-time information from the driver or environment (e.g., video recording of driver and surrounding) are examples of missing information that would help to understand the results better. Complementing the lack of relevant information and the difficulty of guaranteeing that the results of “opportunistic” data have sufficient quality or efficiency, no one can ensure that the results are always associated to the same driver on a single journey. Indeed, [Fitzharris et al. \(2017\)](#) reported the same issue in an experiment with similar technologies.

In addition, the used DMS solution has a lack of information associated to the type of distraction events and the level of drowsiness. Distinguishing the nature of diverting attention could have advantages where different types of distraction might exist and be correlated to different effects on the driver's performance. However, despite the distraction or drowsiness causes, the main effect remains (i.e., safety driving can be affected by the vulnerability to risky situations).

Despite these limitations, interesting conclusions can be pointed out based on the present study's results that may complement other studies. First, and as expected, the driving time is a relevant exposure to the risk variable, which means that by increasing the driving time, the probability of distraction and drowsiness events occurring also increases. Nevertheless, this effect can be highly mitigated if the driver stops during the journey. Although the probabilities of distraction events occurring during the journey is higher than drowsiness events, the impact of increasing the driving time of the journey and stopping during the journey on the probability of drowsiness is higher than the probability of distraction. Overall, the results indicate that the main risk factor of inattention is driving continuously (i.e., without stopping), but it is irrelevant whether the stop is long or short as well as the total time spent on the journey. It should be noted that the journeys' characteristics and the number of distraction and drowsiness events and therefore, the probabilities, may be different from those analyzed in this study if no feedback is provided to the driver.

Regarding the effects of the driver's gender and age on both probabilities, this study reveals that the elderly are more prone to drowsiness. Interestingly, the effect of age on drowsiness has been widely investigated resulting in somewhat different results depending on the analyzed context (simulated vs. real-world environment, sleep deprivation vs. normal sleep, and so on). In contrast to the results obtained, some studies pointed out that the young people are drivers who are more prone to drowsiness events. Moreover, there is a specific drivers' group who did not provide gender information, which is associated to drowsiness and distraction with different trends.

Finally, the study shows that data gathered by a driver monitoring system has the potential of helping to assess risk factors, leading to the main conclusion that non-professional drivers are exposed to risks on long drives, however without any kind of legal control in contrast to professional drivers.

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